React.js with Vite – Manual

1. **Install Visual Studio Code (VS Code)**

https://code.visualstudio.com/sha/download?build=stable&os=win32-x64-user

2. Create a Project Folder – Example: REACT
   Open VS Code.
   Click on File -> Open Folder.
   Create a **new folder** and name it **REACT**.
   Open the **REACT** folder in **VS Code**.

3. **Install Node.js**
   https://nodejs.org/dist/v22.14.0/node-v22.14.0-x64.msi

   Verify the installation by running the following command in the **VS Code terminal**:
   node -v
   npm -v

```
PS E:\REACT\LAB5> node -v
v22.13.1
PS E:\REACT\LAB5> npm -v
11.1.0
PS E:\REACT\LAB5>
```

   Incase if you face **Execution policy error** when you run node -v command

```
npm : File C:\Program Files\nodejs\npm.ps1 cannot be loaded because running
this system. For more information, see about_Execution_Policies at
https:/go.microsoft.com/fwlink/?LinkID=135170.
At line:1 char:1
+ npm run dev
+ ~~~~
    + CategoryInfo          : SecurityError: (:) [], PSSecurityException
    + FullyQualifiedErrorId : UnauthorizedAccess
```

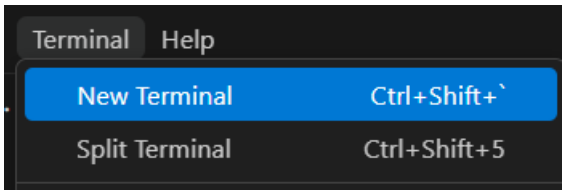
   Open **Windows PowerShell** as **Administrator**.

   Run the following command to set the execution policy:
   Set-ExecutionPolicy -ExecutionPolicy Unrestricted -Scope CurrentUser
   Restart VS Code.

4. Install Vite and Create a React Project

Open VS Code and the **REACT** folder.

Open the terminal



Run this command

npm create vite@latest

If you are running the command for the first time it will ask your permission to install the required packages.

Press Y

```
PS E:\REACT> npm create vite@latest

> npx
> create-vite

? Project name: » vite-project
```

```
√ Project name: ... LAB6
? Package name: » lab6
```

Provide the Project name and Package name.

Package name can be same as project name with lower case letters.

```
? Select a framework: » - Use arrow-keys. Return to submit.
    Vanilla
    Vue
>   React
    Preact
    Lit
    Svelte
    Solid
    Qwik
    Angular
    Others
```

Use arrow keys and select REACT framework and then select JavaScript as the variant.

```
? Select a variant: » - Use arrow-keys. Return to submit.
    TypeScript
    TypeScript + SWC
>   JavaScript
    JavaScript + SWC
    React Router v7 ↗
```

```
√ Project name: ... LAB6
√ Package name: ... lab6
√ Select a framework: » React
√ Select a variant: » JavaScript

Scaffolding project in E:\REACT\LAB6...

Done. Now run:

  cd LAB6
  npm install
  npm run dev

PS E:\REACT>
```

- Run the below commands

cd LAB6
npm install
npm run dev

```
PS E:\REACT>   cd LAB6
>>   npm install
>>   npm run dev
```

```
VITE v6.1.0  ready in 303 ms

➜ Local:   http://localhost:5173/
➜ Network: use --host to expose
➜ press h + enter to show help
```

Open the browser and visit the URL displayed in the terminal (e.g., http://localhost:5173).

To create more projects like LAB2, LAB3, and so on:

Navigate to the REACT folder:

cd ..

Run this command

npm create vite@latest

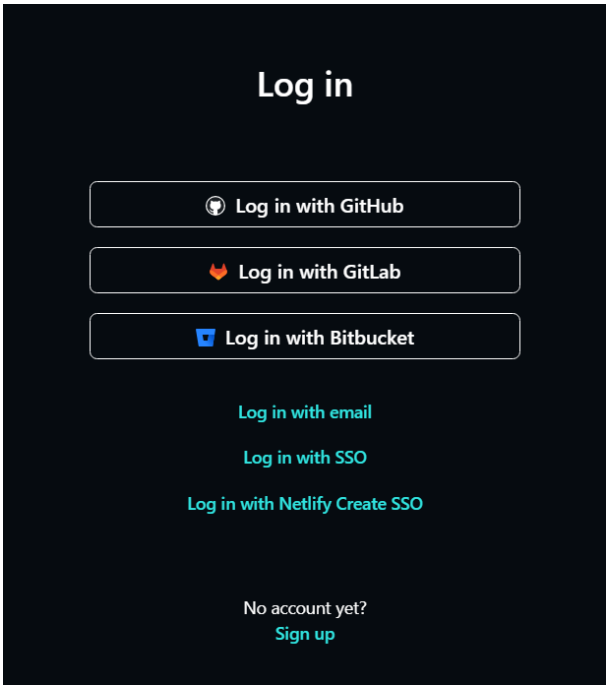
Repeat the same steps as described above to run the application.

Deploying a Static Site

https://vite.dev/guide/static-deploy.html

We are taking our local project and hosting it on a server so others can access it via a URL.

https://www.netlify.com/



Login via email or github

Go inside your program directory using cd command

```
PS E:\REACT> cd lab6
```

 # Install the Netlify CLI

npm install -g netlify-cli

```
PS E:\REACT\lab6> npm install -g netlify-cli
npm warn deprecated glob@8.1.0: Glob versions prior to v9 are no longer supported

changed 1191 packages in 1m

252 packages are looking for funding
  run `npm fund` for details
```

 # Create a new site in Netlify

ntl init

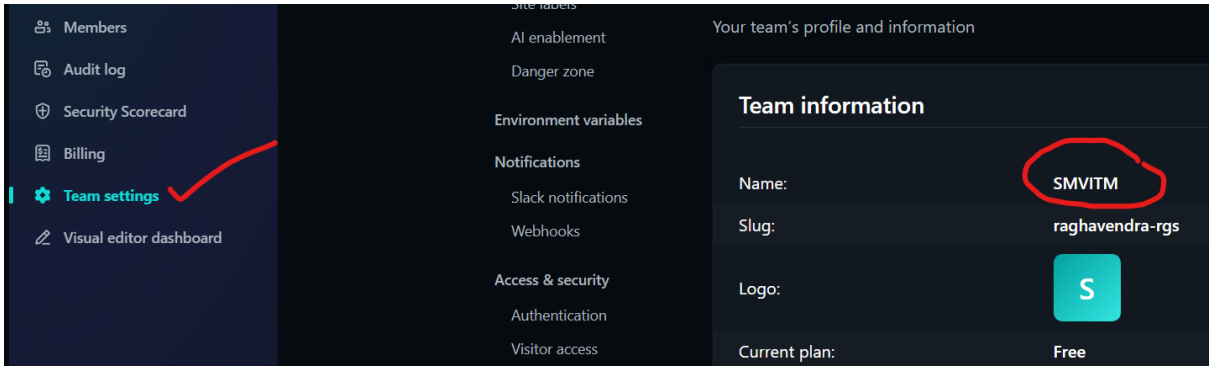
```
PS E:\REACT\lab6> ntl init

No git remote was found, would you like to set one up?

It is recommended that you initialize a site that has a remote repository in GitHub.

This will allow for Netlify Continuous deployment to build branch & PR previews.

For more details on Netlify CI checkout the docs: http://bit.ly/2N0Jhy5

? Do you want to create a Netlify site without a git repository? Yes, create and deploy site manually
? Team: SMVITM
```

Press **Enter Key** to select the first option

here it will display your Netlify team.

Incase if the team's name is not displaying.

Go back to your Netlify account in the browser and change the team settings and provide a name for the team if not done initially.
In the below screenshot you can clearly see the name of the team as **SMVITM**



Once done.

run the **ntl init** command again so as to display the team's name.

```
PS E:\REACT> ntl init

No git remote was found, would you like to set one up?

It is recommended that you initialize a site that has a remote repository in GitHub.

This will allow for Netlify Continuous deployment to build branch & PR previews.

For more details on Netlify CI checkout the docs: http://bit.ly/2N0Jhy5

? Do you want to create a Netlify site without a git repository? Yes, create and deploy site manually
? Team: (Use arrow keys)
> SMVITM
```

SMVITM – TEAM NAME

After you press the enter key it will ask you provide a random site name.

Follow this pattern for the site name: **lab1082** **(lab1 – program name & 082 – USN)**

MANDATORY for all the students to do this as your CIE marks will be awarded based on this for all your lab programs. If not deployed online you will lose your CIE marks unless it's because of some genuine errors during deploying.

```
? Site name (leave blank for a random name; you can change it later): lab1082

Site Created

Admin URL: https://app.netlify.com/sites/lab1082
URL:       https://lab1082.netlify.app
Site ID:   06a52e7a-42e3-4a98-85e7-c7b853426807
```

To deploy to this site. **Run your site build** and then **netlify deploy**

```
PS E:\REACT\lab6> netlify build

Netlify Build
────────────────────────────────────────────────────────────

❯ Version
  @netlify/build 29.58.9

❯ Flags
  accountId: 67a8f89a0e3cc324acd2c255
  dry: false
  offline: false

❯ Current directory
  E:\REACT\LAB6
Build command from Netlify app
────────────────────────────────────────────────────────────

$ npm run build

> lab6@0.0.0 build
> vite build

vite v6.1.0 building for production...
✓ 32 modules transformed.
dist/index.html                  0.46 kB │ gzip:  0.29 kB
dist/assets/react-CHdo91hT.svg   4.13 kB │ gzip:  2.05 kB
dist/assets/index-n_ryQ3BS.css   1.39 kB │ gzip:  0.71 kB
dist/assets/index-Dla6ZdmG.js  186.49 kB │ gzip: 58.97 kB
✓ built in 1.01s

(build.command completed in 2.3s)

Netlify Build Complete
────────────────────────────────────────────────────────────

(Netlify Build completed in 4.1s)
```

Once the build is completed.

Run **netlify deploy** command.

```
PS E:\REACT\lab6> netlify deploy
Please provide a publish directory (e.g. "public" or "dist" or "."):
? Publish directory E:\REACT\lab6\dist
Deploy path: E:\REACT\lab6\dist
Deploying to draft URL...
⠋ Uploading blobs to deploy store...

Netlify Build
────────────────────────────────────────────────────────────

❯ Version
  @netlify/build 29.58.9

❯ Flags
  deployId: 67af5209aa50e32fb3315f3b
  open: false
  prod: false
```

give the directory as **dist**

```
✓ Finished uploading 1 assets
✓ Deploy is live!

Build logs:        https://app.netlify.com/sites/lab1082/deploys/67af5209aa50e32fb3315f3b
Function logs:     https://app.netlify.com/sites/lab1082/logs/functions?scope=deploy:67af5209aa50e32fb3315f3b
Edge function Logs: https://app.netlify.com/sites/lab1082/logs/edge-functions?scope=deployid:67af5209aa50e32fb3315f3b
Website draft URL: https://67af5209aa50e32fb3315f3b--lab1082.netlify.app

If everything looks good on your draft URL, deploy it to your main site URL with the --prod flag.
netlify deploy --prod
```

Run **netlify deploy --prod** command to deploy it.

```
● PS E:\REACT\lab6> netlify deploy --prod
  Please provide a publish directory (e.g. "public" or "dist" or "."):
  ? Publish directory E:\REACT\lab6\dist
  Deploy path: E:\REACT\lab6\dist
  Deploying to main site URL...
  ⠋ Uploading blobs to deploy store...

  Netlify Build
  ────────────────────────────────────────────────────────────

  ❯ Version
    @netlify/build 29.58.9
```

give the directory as **dist**

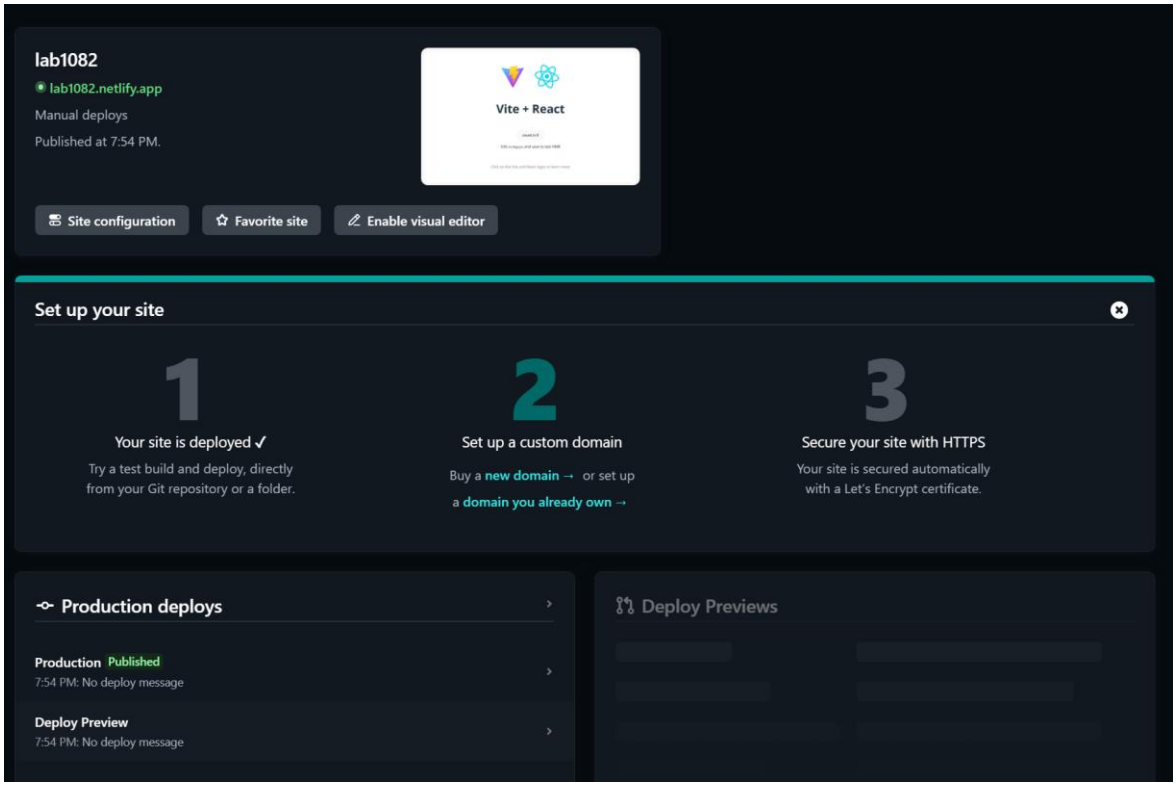
```
❯ Current directory
  E:\REACT\LAB6

❯ Config file
  No config file was defined: using default values.

❯ Context
✓ Finished uploading blobs to deploy store
✓ No cached functions were found
✓ Finished hashing
✓ CDN requesting 0 files
✓ Finished uploading 0 assets
✓ Deploy is live!

Build logs:        https://app.netlify.com/sites/lab1082/deploys/67af522575e12242a7b02a3e
Function logs:     https://app.netlify.com/sites/lab1082/logs/functions
Edge function Logs: https://app.netlify.com/sites/lab1082/logs/edge-functions
Unique deploy URL: https://67af522575e12242a7b02a3e--lab1082.netlify.app
Website URL:       https://lab1082.netlify.app
PS E:\REACT\lab6>
```

Deployment is Completed.

you can visit the website url now to check if it is working or not.

You can ask your friends to test your URL on their devices to check if it is working properly.

You can also cross check the status of the website in your Netlify account



LAB PROGRAM 1:

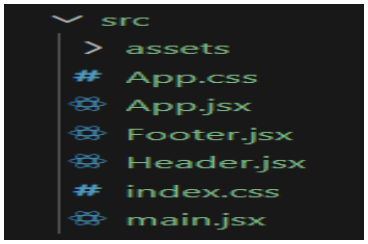
Use create-react-app to set up a new project. Edit the App.jsx file to include a stateful component with useState. Add an input field and a element that displays text based on the input. Dynamically update the content as the user types.

App.jsx

```
import  { useState } from 'react';
import './App.css';

function App() {
  const [text, setText] = useState('');

  return (
    <div className="App" >
      <input
        type="text"
        placeholder="Type something..."
        value={text}
        onChange={(e) => setText(e.target.value)}
      />
      <h1><strong>You Typed: {text}</strong></h1>
    </div>
  );
}
export default App;
```

<u>LAB PROGRAM 2:</u> Develop a React application that demonstrates the use of props to pass data from a parent component to child components. The application should include the parent component named App that serves as the central container for the application. Create two separate child components, Header: Displays the application title or heading. Footer: Displays additional information, such as copyright details or a tagline. Pass data (e.g., title, tagline, or copyright information) from the App component to the Header and Footer components using props. Ensure that the content displayed in the Header and Footer components is dynamically updated based on the data received from the parent component.

In this program you need to create two extra files named Header.jsx and Footer.jsx



App.jsx

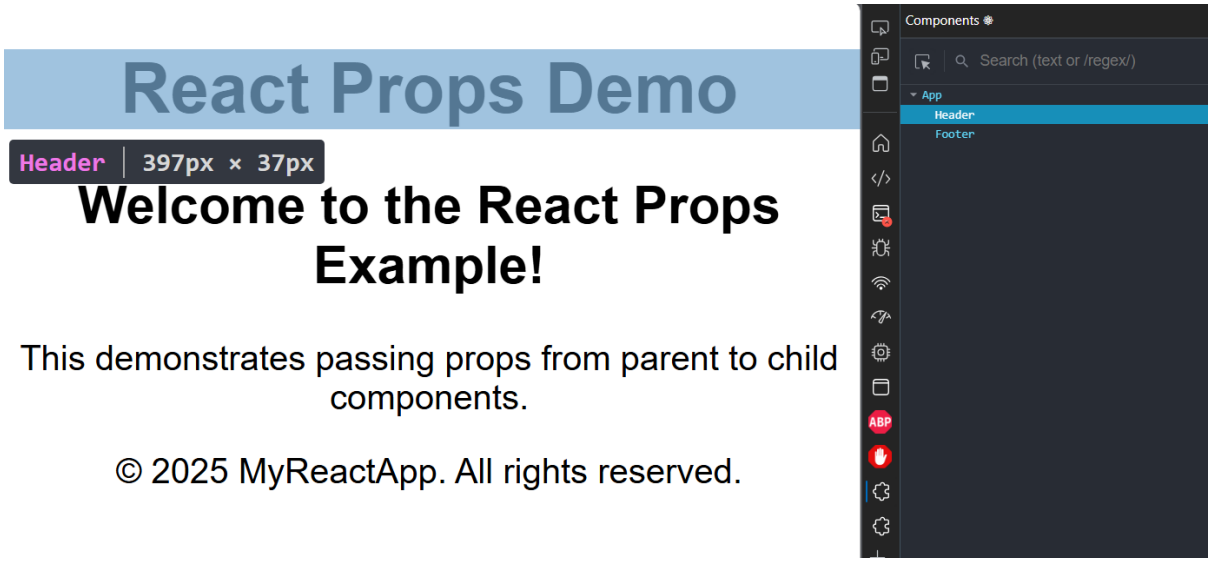
```
import Header from './Header';
import Footer from './Footer';

const App = () => {
  return (
    <div style={{ textAlign: 'center' }}>
      <Header title="React Props Demo" />
      <main>
        <h2>Welcome to the React Props Example!</h2>
        <p>This demonstrates passing props from parent to child components.</p>
      </main>
      <Footer copyright="© 2025 MyReactApp. All rights reserved." />
    </div>
  );
};
export default App;
```

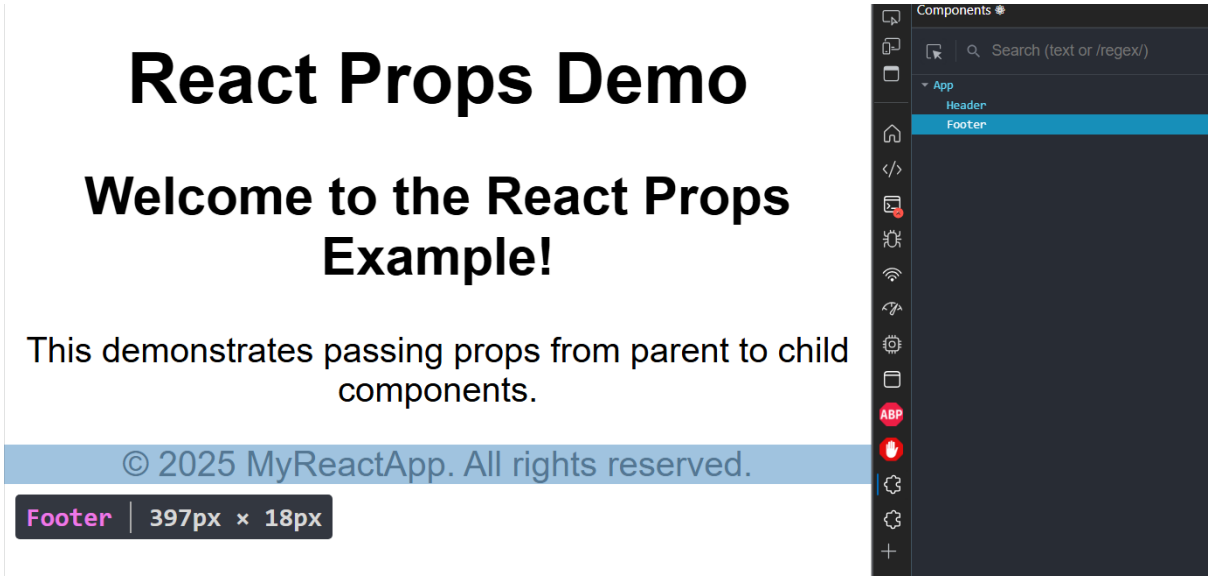
Footer.jsx

```
import PropTypes from 'prop-types';
const Footer = ({ copyright }) => {
  return (
    <footer>
      <p>{copyright}</p>
    </footer>
  );
};
Footer.propTypes = {
  copyright: PropTypes.string.isRequired,
};
export default Footer;
```

Header.jsx

```
import PropTypes from 'prop-types';
const Header = ({ title }) => {
  return (
    <header>
      <h1>{title}</h1>
    </header>
  );
};
Header.propTypes = {
  title: PropTypes.string.isRequired,
};
export default Header;
```





Use React Developer Tools to inspect React components, edit props and state, and identify performance problems.

The easiest way to debug websites built with React is to install the React Developer Tools browser extension. It is available for several popular browsers.

Install for Chrome

Install for Firefox

Install for Edge

Now, if you visit a website built with React, you will see the *Components* and *Profiler* panels when you do inspect element.